



Birla Institute of Technology & Science, Pilani

Pilani Campus

AUGS/AGSR Division

GPIC 2026 | Gujarat Police Innovation Challenge

Module 01 — PRAHARI

Lectures Stakeholder briefing

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Outcomes

- Tell the school-notebook story: cameras already exist, the watchlist is unused
- Name the hybrid (Models 1 + 2 + thin 3) and say why Model 4 is Phase-2
- Trace one frame through capture, analyse(), matcher, alert, CSV
- Separate MEASURED laptop facts from DESIGN TARGET 80k / 5–6 Cr numbers
- State the FRS law: enrolled gallery on Own cameras, never Paldi Circle



Who this briefing is for

- Home Department evaluators, departmental operators, the student team
- Voice: a lecturer naming parts a ten-year-old can follow
- Every number is MEASURED, DESIGN TARGET, or unused (CONJECTURED)
- Page: <https://sentinel.gujarat.gov.in/problems>
- Category 1 student. Lock 07 September 2026, 12:00 IST



The school-notebook analogy

- 26 schools. Each bought a different cupboard for tapes
- Some cupboards keep 7 days. Some keep 15
- Head office has a notebook of stolen bicycles and missing children
- Nobody in the camera room opens that notebook
- That is Gujarat CCTV: 26 departments, analog and IP, 1,000 km
- Watchlists exist (VAHAN, SARTHI, eGujCop, AFIS, NAFIS) and sit unused



What the State already has

- Independent CCTV estates. No statewide camera census
- Cloud NVR in some estates, local storage in others
- Sites: border districts, Valsad, Dahod, Somnath, Jamnagar, Dwarka
- The official goal: use existing infrastructure first
- A new cupboard for every tape is the wrong first move



Four official challenges

- Heterogeneous infrastructure: vendors, VMS, AMC, formats, protocols
- Geographical dispersion of about 1,000 km
- Unified analytics and event handling across onboarded cameras
- New cameras and departments without a redesign

Core goal on the official page: use existing infrastructure to the maximum practical extent.



Reference models (official)

Model	Official name	School picture
1	Registry and GIS	Roll call. No live picture required
2	Unified viewing	One window, many classrooms
3	VMS federation	Post office between cupboards
4	Central VMS	One giant cupboard for the city
H	Hybrid	Combine two or more

- Model 1 is mandatory foundation for 2, 3, and 4



Options we rejected

- Model 1 only: a roll call does not catch a stolen car
- Model 2 only: viewing without a watchlist fails the test case
- Full Model 3 SDK zoo: seven days cannot wrap every vendor
- Model 4 statewide: we cannot host 80,000 recordings (DESIGN TARGET)
- Wild FRS on Paldi Circle: unknown members of the public
- A ministry biometric or vehicle API: we have none. Claiming one is false



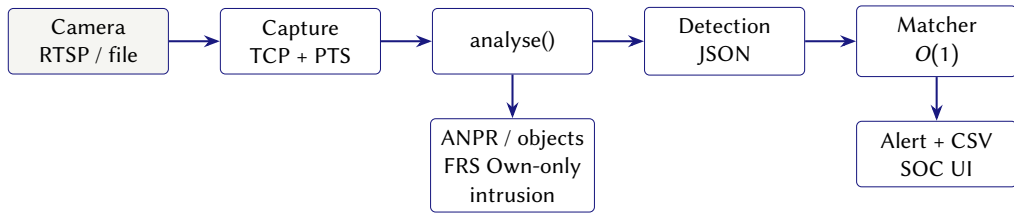
Chosen model

Layer	Official model	PoC
Census, GIS, health, gaps	1	now
Unified viewing + analytics	2	now
Event bus, watchlist, track	thin 3	now
Central VMS recording	4	Phase-2, selected cameras

- A new vendor is one adapter on RTSP / HLS / WHEP / ONVIF
- Detection JSON does not change if the bus later becomes Kafka



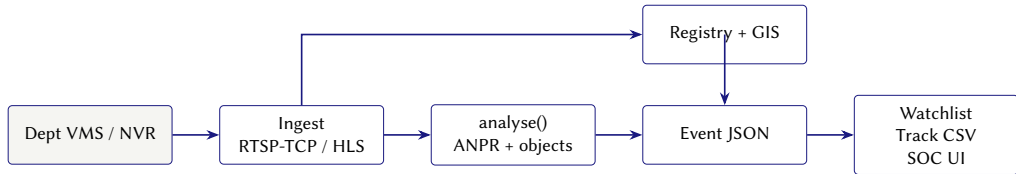
How a picture becomes a red card



- Capture stamps PTS. analyse() samples about 1 fps
- Matcher is $O(1)$. Stolen plate raises CRITICAL. CSV keeps timestamps



Logical architecture



- PoC bus is an in-process WebSocket. The JSON is the contract
- Analytics samples 1 frame per second, not a 25 fps GOP



Official test case

- Onboard about 50 heterogeneous cameras from the Resources page
- Integrate live feeds. Enable monitoring and video analytics
- Identify a designated vehicle registration number across the grid
- Reconstruct route, timestamps, locations. Plot on GIS
- Match detections to a representative watchlist. Raise automated alerts
- Output: working platform, not a mock. CSV / report with timestamps



Live Sentinel contract (03 Sep 2026)

- Portal: `cctv.corp8.cloud` after access-password login
- Catalogue: GET `/cameras.json` returns 30 rows {id, name}
- `/api/ingest` is HTTP 404 on this host
- HLS: `https://cctv.corp8.cloud/<id>/index.m3u8` (cookie + browser UA)
- RTSP: `rtsp://103.250.160.189:8554/stream/<id>` (TCP)
- One second of video takes one second to arrive. There is no file download



Integrator laws in the capture client

- Set `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before `import cv2`
- Event time is PTS (`CAP_PROP_POS_MSEC`), never arrival time, never declared FPS
- Reconnect backoff starts at 2 s and caps at 30 s
- Decoder warnings at join are logged, not fatal
- PTS rewind or gap > 5 s is a scene cut (feeds loop)
- Consume only. Do not `wget /stream/<id>`
- Tests: `tests/test_integrator_laws.py`



Why those laws (ten-year-old version)

- UDP across a firewall arrives as a broken picture that looks like a bad model
- After join, extra frames may arrive fast. Wall-clock time invents flying cars
- Feeds restart. Reconnecting in a tight loop is shouting at a locked door
- The tape loops. At the splice, forget the old object ids
- A downloaded /stream file is a fake dataset



MEASURED: government feed (03 Sep 2026)

- Synced 30 catalogue cameras into the registry (cam01 ...cam30)
- RTSP-TCP probe live: cam04 Paldi Circle, cam01 Chimanbhai Bridge
- First live frame: 03_Data/recordings/first_live_frame.png
- Size 1920×1080, PTS 1080 ms. UDP SETUP 461 then TCP succeeds
- Manifest has no live flag and no lat/lon. Pins stay off the map
- Open tiles from the Cameras table, not from a 0,0 marker



MEASURED: designated vehicle

Time	Camera	Location
06:12	CAM-VAL-001	Valsad NH-48 Toll
07:48	CAM-SUR-001	Surat Adajan Circle
09:05	CAM-AHD-004	Narol–Naroda Hwy
09:22	CAM-AHD-001	SG Highway Junction
09:51	CAM-GNR-003	Koba Circle
10:04	CAM-GNR-001	CH-0 Sector 21 GNR
11:40 IST	cam04	Paldi Circle (live confirm)

- Plate GJ01AB1234, watchlist WL-001 / STOLEN
- Live confirm appends. It does not drop the six seed points



analyse(): one door

- Default engines: anpr, objects
- Own cameras may add faces
- cam plus digits, and any Gov camera, never run FRS
- Godown CAM-FCS-001 also runs person-in-ROI intrusion
- Plate fields stay frozen. New fields are additive
- Ingest writes crops, inserts SQLite, then calls the matcher



ANPR: options

- Train a detector from scratch: no labelled Gujarat set in seven days
- Require GPU YOLO now: GPU count MEASURED 0 on this laptop
- Morphology + Tesseract behind recognize(): CPU, swappable
- Operator confirm: a readable plate cannot fail a live demo
- Confirm writes `source=operator_confirm`. Never labelled ANPR



ANPR: mechanism

- Sampled frame → plate box → Tesseract → Indian normaliser
- Regex after stripping space and hyphen: `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`
- Interface: `recognize(frame) -> {plate, plate_raw, confidence, crop}`
- Missing Tesseract binary: empty plate, no crash
- MEASURED 03 Sep: confirm on cam04, confidence 1.0
- Production swap (same function): YOLO plate detector + PaddleOCR



Objects: options

- YOLOv8n as the only path: extra dependency, tests would skip
- OpenCV DNN with large weights: allowed later, not required for pytest
- Skin-tone blob fallback: a rectangle $\geq 8\%$ of the frame is a person
- Skipping objects: eval area 05 names person and vehicle detection
- Classes kept: person, car, motorcycle, bus, truck, bicycle



Intrusion on a godown

- Food and Civil Supplies watches a warehouse, not a highway
- ROI lives in `cameras.extra_json`
- Person box overlapping the ROI by $\geq 30\%$ of its area fires
- Event: `entity_type=intrusion, CRITICAL`
- Same camera within 120 s increments a counter
- No fourth neural net. Intrusion wraps object boxes



FRS: options we refused

- No FRS: HLD text and eval 05 name recognition
- Wild FRS on Paldi Circle: unknown members of the public
- Live AFIS / NAFIS: we have no ministry API. Claiming one is false
- Enrolled gallery on Own cameras: consented adults or synthetic fixtures
- Human confirm-face exists, same safety net as plate confirm



FRS law (non-negotiable)

- Own-feed may show a gallery match on CAM-OWN-001
- Government CCTV of unknown people never runs FRS
- Tests post a face-like still to cam04 and expect zero person events
- Gallery JSON has no embeddings
- Spoken line: representative person watchlist, not a NAFIS join



Watchlist and alerts

- Representative sources. No live ministry API in the PoC
- $O(1)$ on plate and on face_id / gallery_id
- CRITICAL stolen/wanted/intrusion. HIGH blacklist or person associate
- Same entity + same camera within 120 s increments a counter
- WebSocket /ws/alerts plus 5 s poll fallback
- WL-004 fires without a plate. Person cards show a name, not a blank
- MEASURED: confirm on cam04 raised CRITICAL for GJ01AB1234



Registry, GIS, gaps

- CSV, form, REST, catalogue sync
- Dahod GSRTC is seeded offline on purpose
- Short retention: Food and Civil Supplies 7-day cameras
- Private-permitted mall requires `consent=true`
- `home.viewer` never sees that mall
- Live catalogue has no coordinates. Open `cam04` from the table



Evaluation map A (official areas 1–4)

Official area	PRAHARI now
1 Test case on gov feed	30 cameras onboarded. RTSP-TCP frame. Confirm + CSV on cam04
2 Solution presentation	This deck + notes. Hybrid justified
3 Architecture	HLD. Open adapters. Model 4 not faked
4 Working platform	FastAPI on :8080. Own-feed file + live catalogue



Evaluation map B (official areas 5–7 and bonus)

Official area	PRAHARI now
5 Analytics output	ANPR/confirm + objects CSV + godown intrusion + own-feed gallery FRS
6 Scale / PoC ready	1 fps regional DESIGN TARGET in HLD §5 / §10
7 Completeness	GitHub live. YouTube and Drive still human
Bonus (on camera)	Hybrid, track, objects, intrusion, lawful FRS, RBAC, gaps, WS



Test families (04 Sep 2026)

- 74 passed, 1 skipped (Tesseract binary). `audit_gate.py` PASS
- Integrator: TCP, no CAP_PROP_FPS, scene-cut resets track ids
- ANPR: normaliser, junk drop, confirm $\neq$ source=anpr
- Objects: person blob inserts. YOLO without weights does not crash
- FRS: same gallery matches. cam04 emits zero person events
- Matcher: stolen, blacklist, WL-004 without a plate, 120 s dedupe
- Security: viewer 403 on enroll. Auditor 403 on confirm-face. No RTSP in JSON



Eighty thousand cameras (DESIGN TARGET)

- About 45,000 public-domain cameras run analytics at 1 fps, 720p crop ≈ 80 KB
- Naive centre: $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$
- Five regions (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj): $\approx 720 \text{ MB/s}$ each
- SOC wall pulls 16–64 HLS tiles. Everything else is metadata + crops
- These numbers are design targets, not measured laptop throughput



MEASURED on this laptop (04 Sep 2026)

- Four open file captures succeed. The fifth is rejected
- Mean JPEG from eight own-feed stills: 41 KB
- /api/health p99: 9.4 ms over 50 calls
- GPU count: 0. Regional GPUs stay DESIGN TARGET
- Statewide 80 KB crop remains DESIGN TARGET (ratio $1.9\times$)



Cost of the intelligence plane (DESIGN TARGET)

Item (statewide, per year)	INR
5 regional GPU nodes (3-yr amortised)	1.8–2.4 Cr
Object storage for crops + selected video	0.8–1.2 Cr
SWAN / leased bandwidth top-up	0.6–1.0 Cr
Integration + SOC ops (12 engineers)	1.5 Cr
Total intelligence plane	~5–6 Cr

- This is not the bill for replacing 26 VMS estates



Security in the PoC

- Roles: superadmin, soc_operator, dept_viewer, auditor
- home.viewer sees Home cameras only; private-permitted mall is hidden
- HLS/file tiles use a 60 s HMAC token. Raw RTSP never reaches the browser
- Sentinel cookie stays on the server. The browser never sees it
- Private-permitted onboard requires consent=true
- Audit: onboard, watchlist, ack, report, plate confirm, face confirm



Stack (open source, as required)

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same contract
UI	Leaflet + vanilla JS	React if needed
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream
ANPR	morphology + Tesseract	YOLO + PaddleOCR
Alerts	in-process WebSocket	Redis / Kafka



What is done, what is still human

- Done: platform, HLD, this deck, live catalogue sync, RTSP frame, confirm CSV
- Done: objects, intrusion, lawful FRS with gov refuse, 74 tests
- Done: GitHub github.com/amitduabits/PRAHARI
- Human: own-feed YouTube ≤ 3 min. Gov-feed YouTube ≤ 3 min, no FRS
- Human: Drive Anyone+Viewer for gov_feed_plates.csv
- Human: Tesseract on PATH if OCR should appear on camera
- Stop product edits after the portal receipt on 07 Sep



Demo run (3 minutes)

- Open <http://127.0.0.1:8080>. Login judge
- Operations map: seeded pins, Dahod offline on purpose
- Cameras table: Open tile on cam04 or cam01
- Vehicle Track: reconstruct GJ01AB1234, download CSV
- Alerts: CRITICAL queue, Ack writes audit
- Onboard: Analyse this still, or Operator confirm if OCR misses
- Gaps: short retention on Food and Civil Supplies